



Money Markets & Bond Markets

Team Teaching Financial Markets and Institutions

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The background of the slide is a grayscale photograph of modern architectural structures. On the left, a portion of a white building with rectangular windows is visible. On the right, a taller building with a curved facade and large glass panels is shown, with palm trees in the foreground. A solid dark blue vertical band runs down the center of the image, serving as a backdrop for the text.

01

Money Market Securities



Money Market Securities

Treasury Bills (T-bills)

Short-term securities issued by the U.S. Treasury to finance the budget deficit. They are backed by the federal government, making them virtually free of credit risk and highly liquid.

Commercial Paper

Short-term debt instruments issued by well-known, creditworthy firms to provide liquidity or finance investments in inventory and accounts receivable. They are typically unsecured and offer a slightly higher yield than T-bills due to credit risk.

More Money Market Instruments

Negotiable Certificates of Deposit (NCDs)

Certificates issued by large commercial banks as a short-term source of funds. NCDs have a minimum denomination of \$100,000 and maturities ranging from two weeks to one year.

Repurchase Agreements (Repos)

Agreements where one party sells securities to another with an agreement to repurchase them at a specified date and price. Repos are essentially loans backed by securities.





Federal Funds and Banker's Acceptances



Federal Funds

Enable depository institutions to lend or borrow short-term funds from each other at the federal funds rate, influenced by the supply and demand for funds.



Banker's Acceptances

Indicate that a bank accepts responsibility for a future payment, commonly used for international trade transactions to guarantee payment to exporters.

Treasury Bills in Detail

1

Investors in Treasury Bills

Depository institutions, financial institutions, individuals, and corporations invest in T-bills for liquidity purposes, ensuring easy access to funding.

2

Credit Risk of Treasury Bills

T-bills are backed by the federal government, making them virtually free of credit risk, which is a desirable feature for investors.

3

Liquidity of Treasury Bills

T-bills are highly liquid due to their short maturity and strong secondary market, allowing investors to easily obtain cash by selling them.



Commercial Paper Insights

1

Placement of Commercial Paper

Firms place commercial paper directly with investors or rely on commercial paper dealers, with transaction costs typically around one-eighth of 1 percent.

2

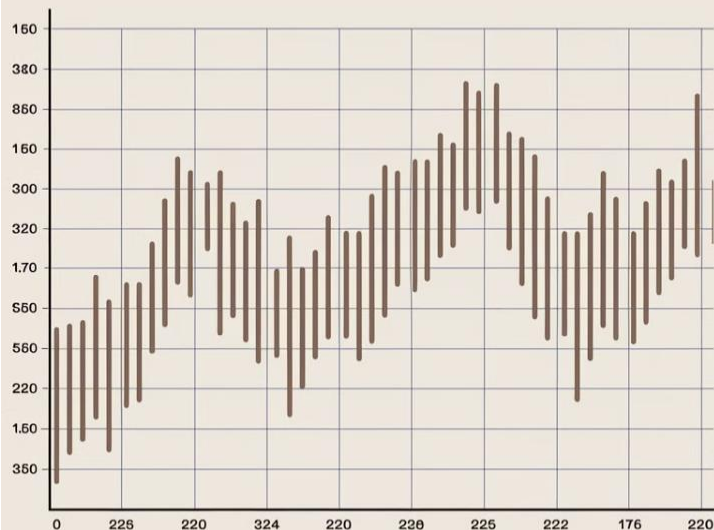
Credit Risk of Commercial Paper

Commercial paper is subject to credit risk, with the risk of default affected by the issuer's financial condition and cash flow.

3

Credit Risk Ratings of Commercial Paper

Credit ratings from agencies like Moody's, S&P, and Fitch help investors assess the credit risk of commercial paper, influencing its marketability.



Negotiable Certificates of Deposit (NCDs)

1

Placement of NCDs

Issuers place NCDs directly, use third-party institutions, or sell to securities dealers, who then resell them to investors.

2

Yield from Investing in NCDs

NCDs provide a return in the form of interest and the difference between the redemption price and the purchase price.

3

NCDs vs. T-Bills

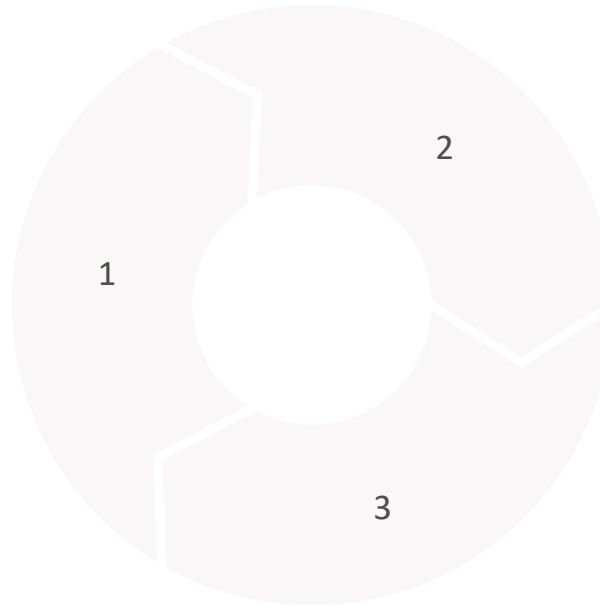
NCDs must offer a slightly higher yield than T-bills with the same maturity to compensate for their lower liquidity and safety levels.



Repurchase Agreements (Repos)

Repo Transactions

Negotiated through a telecommunications network, with dealers and repo brokers acting as intermediaries.



Impact of the Credit Crisis

The credit crisis highlighted that repo market funding requires collateral trusted by investors.

Estimating the Yield

The repo rate is the difference between the initial selling price and the repurchase price, annualized to a 360-day year.

Exhibit 6.6 Summary of Commonly Issued Money Market Securities

SECURITY	ISSUED BY	COMMON INVESTORS	COMMON MATURITIES	SECONDARY MARKET ACTIVITY
Treasury bills	Federal government	Households, firms, and financial institutions	13 weeks, 26 weeks, 1 year	High
Negotiable certificates of deposit (NCDs)	Large banks and savings institutions	Firms	2 weeks to 1 year	Moderate
Commercial paper	Bank holding companies, finance companies, and other companies	Firms	1 day to 270 days	Low
Banker's acceptances	Banks (exporting firms can sell the acceptances at a discount to obtain funds)	Firms	30 days to 270 days	High
Federal funds	Depository institutions	Depository institutions	1 day to 7 days	Nonexistent
Repurchase agreements	Firms and financial institutions	Firms and financial institutions	1 day to 15 days	Nonexistent

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02

Institutional Use of Money Markets



Institutional Use of Money Markets

Exhibit 6.8 Institutional Use of Money Markets

TYPE OF FINANCIAL INSTITUTION	PARTICIPATION IN THE MONEY MARKETS
Commercial banks and savings institutions	• Bank holding companies issue commercial paper. • Some banks and savings institutions issue NCDs, borrow or lend funds in the federal funds market, engage in repurchase agreements, and purchase T-bills. • Commercial banks create banker's acceptances. • Commercial banks provide backup lines of credit to corporations that issue commercial paper.
Finance companies	• Issue large amounts of commercial paper.
Money market mutual funds	• Use proceeds from shares sold to invest in T-bills, commercial paper, NCDs, repurchase agreements, and banker's acceptances.
Insurance companies	• May maintain a portion of their investment portfolio as money market securities for liquidity.
Pension funds	• May maintain a portion of their investment portfolio as money market securities that may be liquidated when portfolio managers desire to increase their investment in bonds or stocks.

03

Globalization of Money Markets



Globalization of Money Markets



1

International Interbank Market

Facilitates the transfer of funds between banks with excess funds and those with deficient funds.

2

Eurodollar Securities

Dollar deposits in non-U.S. banks, used in Eurodollar CDs, Euronotes, and Euro-commercial paper.

Performance of Foreign Securities

Measured by the effective yield, adjusted for the exchange rate effect.

$$Y_f = \frac{SP_f - PP_f}{PP_f}$$

where

SP_f = selling price of the foreign money market security in the foreign currency
 PP_f = purchase price of the foreign money market security in the foreign currency

The exchange rate effect (denoted as $\% \Delta S$) measures the percentage change in the spot exchange rate (in dollars) from the time the foreign currency was obtained to invest in the foreign money market security until the time the security was sold and the foreign currency was converted into the investor's home currency. Thus, the effective yield is

$$Y_e = (1 + Y_f) \times (1 + \% \Delta S) - 1$$



Key Takeaways

Money markets are essential for short-term financing and liquidity management. Key instruments include T-bills, commercial paper, NCDs, repos, federal funds, and banker's acceptances. Financial institutions use these markets to manage liquidity and obtain cash. Globalization has integrated money markets, but investments in foreign securities are subject to exchange rate risk.

04

Background on Bonds



Bond Markets

This slide provides an overview of bond markets, including key players, influencing factors, bond types, institutional participation, macroeconomic drivers, and the market's role in finance.

- Dynamics of bond markets, **including insights from the Indonesian bond market (SUN).**
- Key institutional players in bond markets.
- Factors influencing bond yields, **with examples from Indonesian government and corporate bonds.**
- Characteristics of various bond types.
- Participation of commercial banks, finance companies, and mutual funds in bond markets, **with specific reference to their roles in the Indonesian market.**
- Macroeconomic and firm-specific factors that drive bond prices and yields, **considering Indonesian market dynamics.**
- Role of bond markets in the broader financial landscape, especially regarding institutional investment strategies, **with Indonesian market case studies.**



Bond Issuance and Repayment Process



Bond Issuance

Government or corporation issues bonds

Investors buy the bonds

Investor Purchase



Coupon Payment Obligation

Issuer must pay interest

Evaluate future cash flows

Cash Flow Assessment



Coupon Payment Fulfillment

Issuer pays coupon to investors

Issuer must repay principal

Principal Repayment Obligation



Principal Repayment Fulfillment

Issuer repays principal at maturity

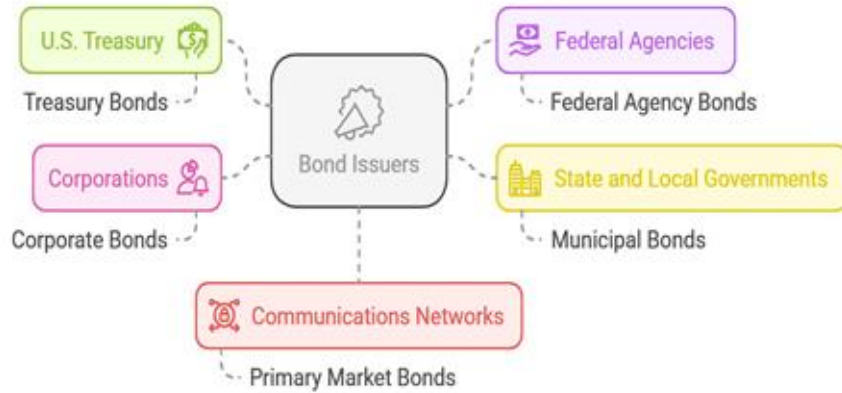
1. Understanding Bond Markets

Bonds are long-term debt securities issued by government agencies or corporations. Most bonds mature in 10–30 years. Bond issuers must pay interest (or coupon) payments annually or semiannually and the par value (principal) at maturity. An issuer must demonstrate that its future cash flows will cover bondholder coupon and principal payments. Investors will only buy questionable-repayment bonds if the expected return outweighs the risk.

1. Understanding Bond Markets



Bond Issuance and Management Structure



Bonds are often categorized by issuer. The U.S. Treasury issues Treasury bonds, federal agencies issue federal agency bonds, state and local governments issue municipal bonds, and corporations issue corporate bonds. A communications network issues primary market bonds. Registered bonds require the issuer to keep track of who owns them so coupon payments can be sent automatically. Financial institutions pay coupons by debiting the issuer's account and crediting the bond owner's account.

Bond Market Financial Flow



Exhibit 7.1 shows how bond markets move money. United States Treasury bonds fund deficit spending on government programs. Federal agencies buy mortgages from financial institutions using bond proceeds. Thus, they indirectly finance home purchases. Companies use bond proceeds to expand. Bond markets boost economic growth by allowing the U.S. government, federal agencies, and corporations to spend more.

1.a. Institutional Participation in Bond Markets



Commercial Banks & S&Ls

Invest in government and corporate bonds for liquidity and income. Issue municipal bonds for local projects and bonds for their capital needs.

Indonesia: Banks actively invest in Indonesian government bonds (SUN) for stable returns and liquidity management. For example, Bank Mandiri and BCA are key players in the SUN market.



Finance Companies

Issue bonds to fund lending in auto, personal, and business sectors. Bond maturities are varied to align with asset terms.

Indonesia: Finance companies in Indonesia issue bonds (obligasi) to diversify funding sources beyond bank loans, supporting growth in consumer finance. Adira Finance and BFI Finance are active issuers in the Indonesian bond market.



Mutual Funds

Offer diversified bond portfolios, from government securities to corporate bonds, catering to varied risk appetites and investment goals.

Indonesia: Indonesian mutual funds offer bond funds (reksa dana pendapatan tetap) focusing on government and corporate bonds, attracting both retail and institutional investors. Manulife and Schroder are examples of fund managers offering popular bond funds in Indonesia.

1.a. Institutional Participation in Bond Markets

The Role of Brokerage and Investment Banking Firms



Brokerage Firms

Facilitate bond trading by matching buyers and sellers, ensuring market liquidity and price discovery.

Indonesia: Indonesian brokerage firms play a key role in trading government bonds (SUN) and corporate bonds, contributing to market efficiency.



Investment Banking Firms

Underwrite new bond issuances for entities, assuming risk or acting on a best-efforts basis.

Indonesia: Investment banks in Indonesia underwrite new bond issuances (emisi obligasi) for corporations and government entities, supporting infrastructure and corporate growth.



Insurance Companies & Pension Funds

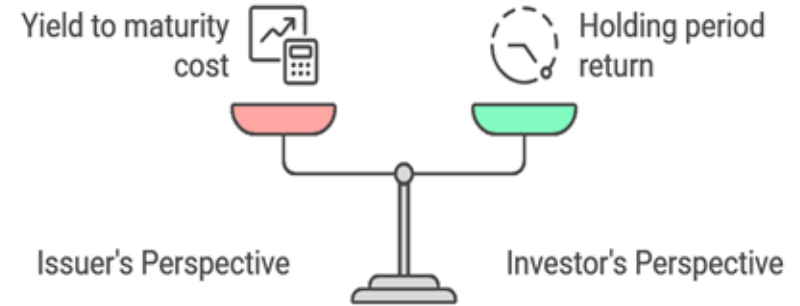
Acquire bonds as long-term investments, seeking stable income to meet obligations.

Indonesia: Insurance companies and pension funds in Indonesia invest in government and corporate bonds to secure long-term returns for policyholders and retirees.

1 b. Bond Yields

Bond yields reflect anticipated return, influenced by coupon rate, market price, and maturity. Yield to maturity (YTM) shows total return if held to maturity, considering market price, face value, coupon rate, and time to maturity. Premium bonds have lower YTM than coupon, discount bonds have higher YTM.

- Bond yields are a critical metric for investors, reflecting the anticipated return on a bond investment. These yields are influenced by several factors, including the bond's stated coupon rate, its current market price, and the time remaining until its maturity date. Bond yields are not static; they fluctuate daily based on market conditions and vary across different maturities, providing insight into market expectations for future interest rate movements. For example, Indonesian Government bonds (Surat Utang Negara or SUN) yields are closely watched as indicators of Indonesia's economic stability.
- The yield to maturity (YTM) offers a comprehensive view of the total return an investor can expect if the bond is held until maturity. Expressed as an annualized rate, YTM takes into account the bond's current market price, its face value (par value), the coupon interest rate, and the remaining time to maturity. In Indonesia, the YTM for corporate bonds can be significantly influenced by the credit rating of the issuing company and prevailing market sentiment.
- For example, a bond trading at a premium will have a YTM lower than its coupon rate, while a bond trading at a discount will have a higher YTM. In the Indonesian market, if Bank Indonesia (BI) is expected to cut interest rates, bonds, including SUN, may trade at a premium, lowering the YTM relative to the coupon rate. Conversely, uncertainty regarding currency stability can cause bonds to trade at a discount, increasing the YTM.



Comparing Bond Yields from Issuer and Investor Perspectives

Understanding Bond Yields: Types and Implications



US Bond yields from several reliable sources:

- Bloomberg:** They provide detailed information on US Treasury yields, including government bonds, municipal bonds, and interest rates.
- Yahoo Finance:** This platform offers comprehensive data on US Treasury bonds, including prices, changes, trading volume, and daily charts.
- Investing.com:** They provide a yield curve for US government bonds, showing yields for various maturities.

Indonesia bond yields from several reliable sources:

- Investing.com:** Provide detailed information on Indonesia's 10-year bond yield, including historical data, charts, and analysis.
- Trading Economics:** This platform offers data on Indonesia's government bond yields, including historical data and news.
- World Government Bonds:** They provide a yield curve for Indonesia government bonds, showing yields for various maturities.



05

Treasury and Federal Agency Bonds

2. Treasury and Federal Agency Bonds

The U.S. government, like many other national governments, commonly spends more money than it receives from taxes. Under these conditions, it needs to borrow funds to cover the difference between what it wants to spend versus what it receives.

To facilitate its fiscal policy, the U.S. Treasury issues Treasury notes and Treasury bonds to finance federal government expenditures.

The Treasury pays a yield to investors that reflects the risk-free rate, as it is presumed that the Treasury will not default on its payments.

Treasury notes and bonds are free from credit (default) risk; they enable the Treasury to borrow funds at a relatively low cost. However, additional borrowing by the U.S. government might potentially cause investors to worry about the Treasury's ability to cover its debt payments. The minimum denomination for Treasury notes and bonds is now \$100.

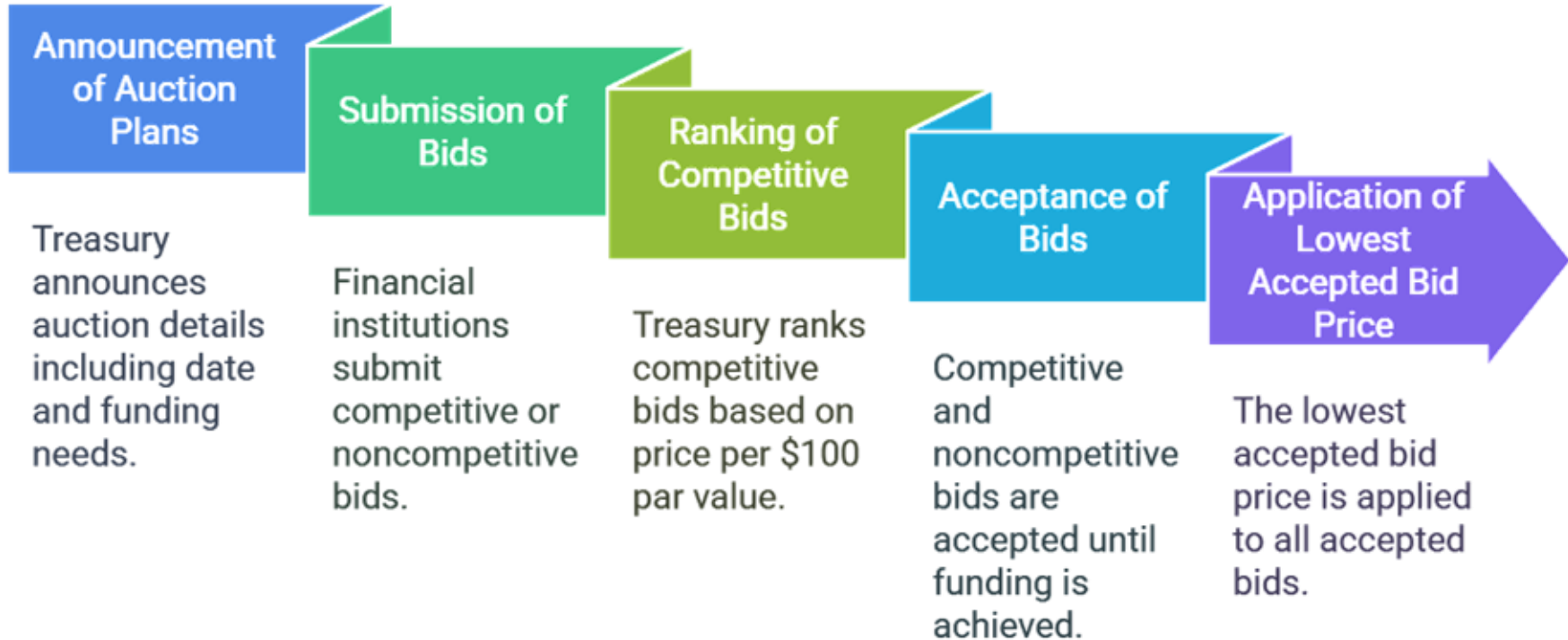
The key difference between a note and a bond is that note maturities are less than 10 years whereas bond maturities are 10 years or more.



2. Treasury and Federal Agency Bonds



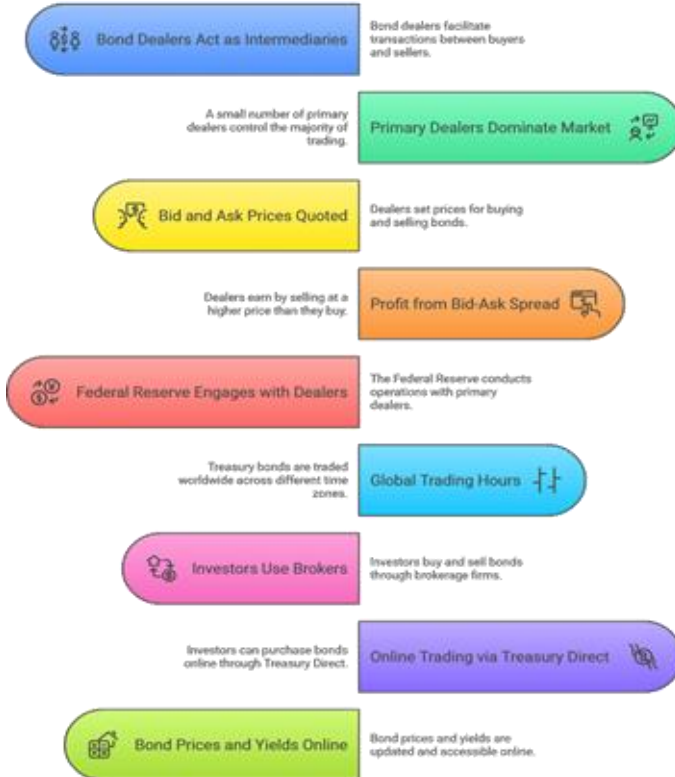
Treasury Bond Auction Process



2. Treasury and Federal Agency Bonds



Treasury and Federal Agency Bonds



Online Trading Investors can also buy bonds through the Treasury Direct program (www.treasurydirect.gov) and have the Treasury deduct the purchase amount from their bank account. When their Treasury bonds mature, they can reinvest the proceeds into newly issued Treasury bonds.

Online Quotations Treasury bond prices are accessible online at the Markets Data Center (www.wsj.com). This website provides the spread between the bid and the ask (offer) prices for various maturities. Treasury bond yields are accessible online at various websites (for example, www.bloomberg.com and www.finance.yahoo.com). The yields are updated daily and are given for several different maturities.

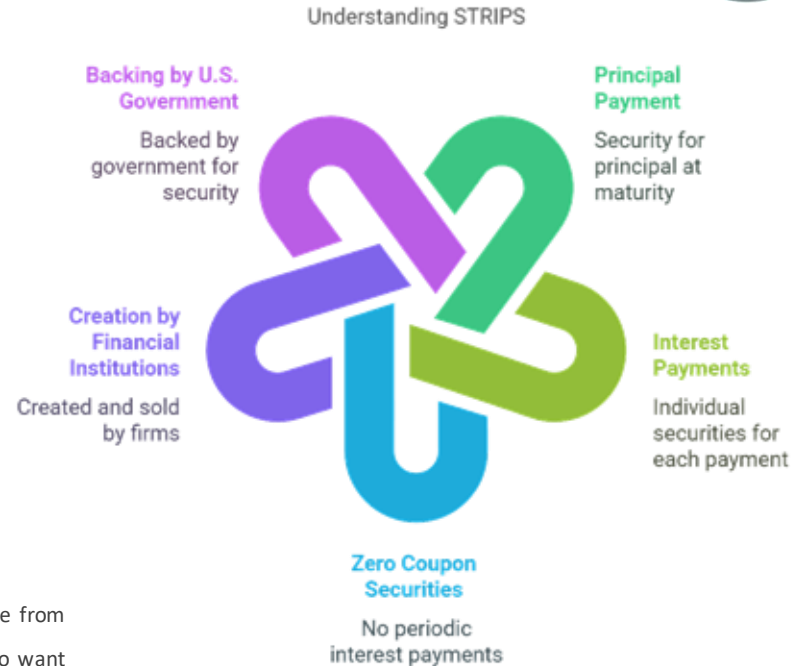
2. Treasury and Federal Agency Bonds



Stripped Treasury Bonds (STRIPS)

STRIPS break Treasury bonds into individual zero-coupon securities, offering flexibility and predictable returns for savings plans.

- STRIPS, or Separate Trading of Registered Interest and Principal Securities, divide a Treasury bond's coupon payments and principal into separate zero-coupon securities. A 10-year Treasury note can become 21 components: 20 coupon payments and a principal payment. Although direct stripping isn't common for Indonesian government bonds (Surat Utang Negara, SUN), similar concepts exist.
- Each component is sold separately for targeted savings or pension fund management. Investors can buy specific coupon payment dates or the final principal repayment. Zero-coupon bonds in Indonesia serve similar purposes, especially in sharia-compliant investments.
- These zero-coupon securities can be traded before maturity, providing flexibility. Returns come from the difference between the purchase price and face value at maturity, attracting investors who want predictable returns without reinvestment risk. In Indonesia, investors can selectively buy bonds with specific maturity dates to align with financial goals, influenced by Bank Indonesia's interest rate policies.





2. Treasury and Federal Agency Bonds

Inflation-Indexed Treasury Bonds (TIPS)

TIPS protect investors from inflation by adjusting the bond's principal based on the Consumer Price Index (CPI) and paying interest on the adjusted principal, ensuring a return above inflation.

- **Principal Adjustment Example (Indonesia):** In Indonesia, inflation-indexed bonds adjust the principal based on the Indonesian CPI. For example, if you invest IDR 10,000,000 in an Indonesian TIPS bond and the CPI rises by 3%, the principal is adjusted to IDR 10,300,000.
- **Coupon Payments Example (Indonesia):** Indonesian TIPS pay interest based on a fixed rate applied to the inflation-adjusted principal. If the TIPS bond has a 2% fixed rate, the interest payment on the adjusted principal (IDR 10,300,000) would be IDR 206,000, ensuring a return above inflation.

2. Treasury and Federal Agency Bonds



Savings Bonds



Overview

Savings bonds are an accessible, tax-advantaged investment option with fixed or inflation-adjusted interest rates. Savings bonds are issued by the Treasury, but they can be purchased from many financial institutions. They are attractive to small investors because they can be purchased with as little as \$25. Larger denominations are also available

Key Features in Indonesia

- Equivalents in Indonesia start at relatively low denominations.
- Retail ORI (Obligasi Ritel Indonesia) bonds offer fixed interest rates.
- Sukuk Tabungan (ST) bonds are Sharia-compliant, offering returns based on profit-sharing.
- Subject to Indonesian tax laws; consult local financial advisors.
- Purchased through banks and online platforms in Indonesia.

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06

Municipal Bonds



3. Municipal Bonds

- **Issuance:** State and local governments issue municipal bonds to fund essential public projects. These can include constructing new schools, improving roadways, or upgrading water and sewer systems. For example, a city might issue bonds to finance a new light rail project to improve public transportation. **In Indonesia, while municipal bonds are less common, similar financing needs are often met through regional government bonds (obligasi daerah), such as those issued by the Jakarta Provincial Government to fund infrastructure projects like the MRT (Mass Rapid Transit) system.**
- **Denomination:** Municipal bonds generally have a minimum denomination of \$5,000, although some may offer bonds in increments as low as \$1,000. This relatively low entry point makes them accessible to a broad range of investors, including individuals looking to diversify their fixed income portfolio. **In Indonesia, retail bonds (Obligasi Ritel Indonesia - ORI) are often available in denominations starting from IDR 1 million (approximately \$70 USD), making them accessible to individual investors.**
- **Secondary Market:** Municipal bonds can be resold on the secondary market, offering liquidity to investors. However, the market is less active than that for Treasury bonds, potentially leading to wider bid-ask spreads. Investors should be aware of market conditions and transaction costs when considering selling bonds before maturity. **The secondary market for Indonesian government bonds is relatively liquid, but liquidity can vary depending on the specific bond and market conditions. Investors can trade bonds on the Indonesia Stock Exchange (IDX) or through over-the-counter (OTC) transactions.**



Tax Advantages of Municipal Bonds in Indonesia

Certain Indonesian government bonds (SUN) offer tax-exempt interest, particularly attractive in higher income brackets. Calculate the tax-equivalent yield to compare effectively with taxable investments.

- Certain Indonesian government bonds (Surat Utang Negara, or SUN) offer a tax advantage: the interest earned may be exempt from income taxes, depending on the specific bond and regulations. This is appealing to individuals in higher tax brackets looking to optimize their tax liabilities within the Indonesian context.
- To effectively compare these Indonesian government bonds to taxable alternatives, calculate the tax-equivalent yield. For example, if a tax-exempt SUN offers a 6% yield and you're in the 30% tax bracket, a taxable investment would need to yield approximately 8.57% to match the after-tax return. This calculation highlights the value of the tax exemption in Indonesia.

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07

Corporate Bonds



4. Corporate Bonds: Funding Business Growth



Corporate bonds are issued by companies to fund growth, offering potentially higher interest rates compared to government bonds. Companies, including Indonesian firms, issue corporate bonds to fund operations, acquisitions, or R&D.

Corporate bonds can be placed with investors through a public offering or a private placement.

Timing

The yield that corporations pay to investors on the bonds is highly dependent on the market interest rates at the time of the placement. Some corporations attempt to time their bond offerings when market interest rates are low, so that their cost of financing with bonds will be low.

Private Placement

Some corporate bonds are privately placed rather than sold in a public offering.

Credit Risk of Corporate Bonds

Corporate bonds are subject to the risk of default, and the yield paid by corporations that issue bonds contains a risk premium to reflect the credit risk. The general level of defaults on corporate bonds is a function of economic conditions.

Bond Ratings as a Measure of Credit Risk When corporations issue bonds, they FINANCIAL REFORM hire rating agencies to rate their bonds.

Corporate bonds that receive higher ratings can be placed at higher prices (lower yields) because they are perceived to have lower credit risk.

As the preceding example described, Apple and Google were able to issue bonds at a yield almost as low as the yield on Treasury securities.

Corporate bonds that are perceived to have very high risk are referred to as **junk bonds**. The primary investors in junk bonds are mutual funds, life insurance companies, and pension funds.

Corporate bonds may offer higher interest rates than Indonesian government bonds (SUN), reflecting their higher risk profile.



Junk Bonds: High Risk, High Reward

Junk bonds, or high-yield bonds, offer potentially greater returns but come with a higher risk of default. Investors, especially in markets like Indonesia, should carefully weigh their risk tolerance.

Rating and Risk

Typically rated BB or lower by S&P (Ba or lower by Moody's), indicating a higher default risk. Indonesian high-yield bonds may offer yields above government bonds to compensate. Bonds from Indonesian property developers with high debt could be considered junk bonds.

Issuers

Companies with high debt, those in rapidly changing industries (tech, biotech), or undergoing restructuring. Could include infrastructure projects with high initial debt or Indonesian telecom companies facing rapid technological change, such as toll road projects or Jakarta tech startups.

Investors and Considerations

High-yield funds and hedge funds are common investors.

Individuals should carefully consider risk tolerance due to sensitivity to economic downturns and potential losses, especially during economic uncertainty in Indonesia.

Monitor financial health of Indonesian companies and consult advisors.



4. Corporate Bonds: Funding Business Growth

Corporate bonds are issued by companies to fund growth, offering potentially higher interest rates compared to government bonds.

- Companies, including Indonesian firms, issue corporate bonds to fund operations, acquisitions, or R&D.
- These bonds are subject to Indonesian taxes, influencing their after-tax yield.
- Corporate bonds may offer higher interest rates than Indonesian government bonds (SUN), reflecting their higher risk profile.
- Issuing these bonds involves underwriters who may purchase the bonds directly or facilitate their sale to institutional investors.
- Corporate bond ratings from Pefindo are crucial for attracting larger investors seeking to manage risk and returns in the Indonesian market.



Bond ratings from agencies like Moody's and S&P assess creditworthiness, impacting risk and bond prices. Ratings guide institutional and individual investors, with downgrades potentially leading to price drops.



Bond Ratings as credit measured risk

- Bond ratings, provided by agencies like Moody's and Standard & Poor's, assess the creditworthiness of corporate bonds.
- For example, a bond rated AAA by S&P indicates a very low risk of default, whereas a BB rating suggests a higher risk, potentially aligning with "junk bond" status. For instance, Indonesian government bonds are often rated based on the country's economic stability and fiscal policies.
- Institutional investors, such as pension funds, rely heavily on these ratings for compliance and risk management. Indonesian pension funds also consider bond ratings when investing in domestic corporate bonds.
- Individual investors can also use these ratings to gauge the risk associated with different corporate bonds. For example, monitoring the ratings of bonds issued by Indonesian corporations can help investors assess potential risks.
- A change in rating, such as a downgrade from A to BBB, can lead to a significant drop in the bond's price. Similarly, a downgrade in Indonesia's sovereign credit rating can impact the prices of bonds issued by Indonesian companies.
- In Indonesia, Pefindo is a key agency providing bond ratings, offering insights into the creditworthiness of Indonesian corporate bonds.



Callable Bonds: An Advantage for Issuers

Advantage for Issuers

Callable bonds allow issuers to redeem bonds early if interest rates fall, reducing borrowing costs. For example, MegaCorp issues callable bonds at 6%; rates fall to 4%. MegaCorp issues new bonds at 4% to retire the 6% bonds. This lowers MegaCorp's borrowing costs, improving its finances. In the US, many corporations used callable bonds in the 1980s to refinance debt at lower rates during periods of economic expansion.

Callable bonds allow issuers to redeem bonds before maturity at a set price. Consider a similar scenario with Indonesian state-owned enterprises (BUMN) issuing callable bonds. This also applies to Indonesian companies looking to optimize their debt.

Risk for Investors

Investors face the risk of early redemption and reinvesting at potentially lower rates. Indonesian bondholders face the same risk when companies call back bonds early due to favorable market conditions. Similarly, US investors in callable bonds must consider this reinvestment risk, particularly when interest rates are volatile.



Using Bonds to Finance a Leveraged Buyout

Bonds are crucial for leveraged buyouts (LBOs), where acquired firms' assets secure high-yield bonds. Increased debt can cause distress, but successful LBOs can yield high returns from efficiency, realignment, or sale at a higher valuation.

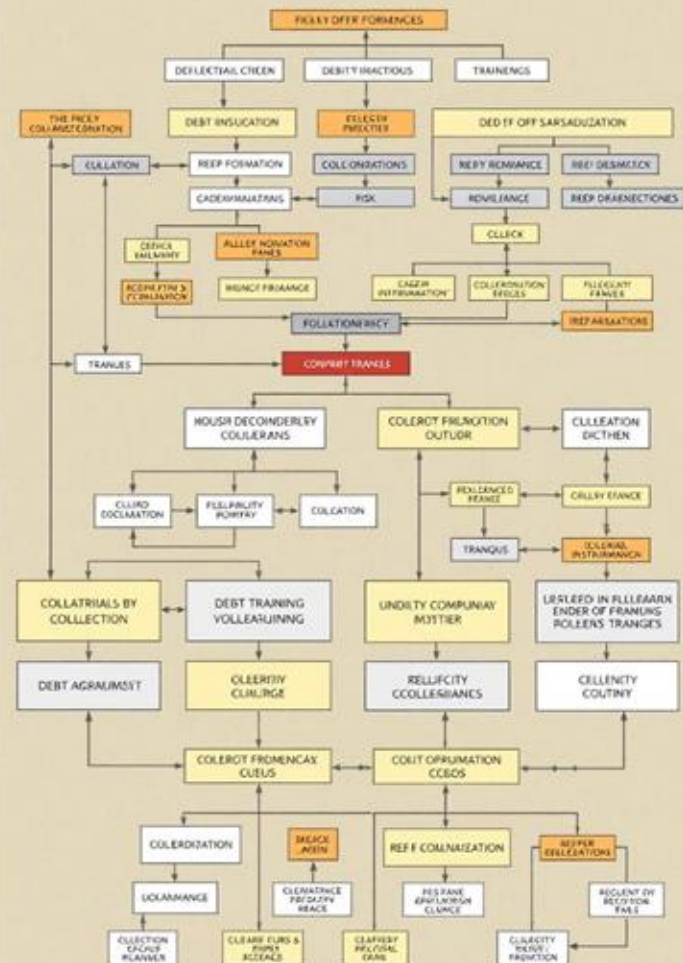
- **Key Tool in LBOs:** Bonds are a key tool in leveraged buyouts (LBOs), where a company is acquired primarily with borrowed funds.
- **High-Yield Bonds:** Typically, the acquiring firm issues high-yield bonds to finance the majority of the purchase price. For example, a private equity firm might use bond financing to acquire an Indonesian telecom company.
- **Collateral:** The acquired company's assets and cash flows serve as collateral for the debt. For instance, assets of a manufacturing company in Indonesia could be used.
- **Increased Debt Burden Risks:** This increased debt burden can pose significant risks, potentially leading to financial distress if the company fails to meet its debt obligations. This is a risk for Indonesian companies as well.
- **Potential Returns:** However, successful LBOs can generate substantial returns for investors, who benefit from increased efficiency, strategic realignment, or eventual sale of the company at a higher valuation.
- **Scenario:** Consider a scenario where an LBO increases an Indonesian company's value by 20% within five years, allowing the private equity firm to sell it for a significant profit after repaying the bonds.

Collateralized Debt Obligations (CDOs)

CDOs are complex investments bundling debts into tranches of varying risk. Popular from 2004-2007, they faced criticism for opacity and their role in the 2008 financial crisis due to subprime mortgage reliance and inflated ratings.

- Collateralized debt obligations (CDOs) are complex investments that bundle together various types of debt, such as residential mortgages, commercial loans, and corporate bonds. Indonesian CDOs might include sovereign bonds issued by the Indonesian government or corporate bonds from Indonesian companies.
- These debts are then categorized into tranches based on their risk profile: senior tranches (rated AAA), mezzanine tranches (rated BBB), and equity tranches (unrated). For example, a senior tranche might consist of Indonesian government bonds, considered relatively safe.
- CDOs gained traction between 2004 and 2007 as investors sought higher yields in a low-interest-rate environment. In Indonesia, this period saw increased issuance of CDOs backed by a mix of local and international debt.
- CDOs were also criticized for their opacity and complexity, contributing significantly to the 2008 financial crisis. The lack of transparency in CDOs containing Indonesian assets also posed challenges for investors.
- The over reliance on subprime mortgages within CDOs led to widespread defaults when the housing bubble burst. While Indonesian mortgages were not directly involved, the global impact affected Indonesian markets as well.
- Credit rating agencies like Moody's and Standard & Poor's faced scrutiny for assigning inflated ratings to CDOs, misleading investors about their true risk. Similar concerns arose regarding the ratings of CDOs with exposure to Indonesian debt.

COLLATERAL DEBT OBLIGATION (CDO) + CMs



08

Globalization of Bond and Loan Markets



The CDS Story: Subprime Mortgages and the 2008 Financial Crisis (CREDIT CRISIS)

Collateralized Debt Obligations (CDOs) played a significant role in the 2008 financial crisis.

These complex investments bundled together various types of debt, including subprime mortgages. The over-reliance on these subprime mortgages, combined with inflated credit ratings, created a recipe for disaster.

When the housing bubble burst, widespread defaults on subprime mortgages triggered massive losses within CDOs. This led to a cascading effect, as investors lost confidence in these investments. The complexity and lack of transparency surrounding CDOs made it difficult to assess the true risk, further exacerbating the crisis.

Check out this video!

[Understanding CDSs and the Financial Crisis.](#)



Investing in foreign bonds allows diversification, accessing markets like emerging economies or stable European nations for higher returns, while currency exchange rates pose a risk.

Investing in Foreign Bonds

Europe

Investing in European bonds offers exposure to a mix of developed economies and the Euro currency. The European Central Bank's policies influence bond yields, and investors often consider sovereign debt from countries like Germany or France for stability.

United States

The U.S. bond market is the largest in the world, providing diverse options from government securities to corporate bonds. Interest rates set by the Federal Reserve significantly impact bond values, and the U.S. dollar's strength affects international investors.

Indonesia

Indonesian bonds, including Surat Berharga Negara (SBN), can offer higher yields compared to developed markets. The Indonesian Rupiah's volatility and Bank Indonesia's monetary policy are key factors. Investing in Indonesian corporate bonds, like those from Telkom Indonesia, can also be attractive.





Sovereign Debt

Sovereign debt arises when governments borrow to cover expenses. Credit ratings affect borrowing costs, and crises can disrupt global markets. International bodies like the IMF offer aid with conditions.

- Sovereign debt occurs when national governments borrow money to finance expenses. For example, the Indonesian government issues bonds in Rupiah and foreign currencies to fund infrastructure projects and budget deficits.
- Credit rating agencies evaluate the creditworthiness of governments. Factors like GDP growth, inflation rates, political stability, and foreign exchange reserves are considered. For instance, agencies assess Indonesia's fiscal policies and economic resilience when assigning its sovereign credit rating.
- Sovereign debt crises can severely impact global financial markets. These crises can lead to increased borrowing costs for other nations and decreased investor confidence. Indonesia experienced financial challenges during the Asian Financial Crisis in the late 1990s, impacting its sovereign debt and requiring international assistance.
- International organizations like the IMF often provide financial assistance to countries facing debt crises. However, this assistance typically comes with conditions, such as economic reforms. For example, Indonesia has worked with the IMF on various economic programs to maintain financial stability and manage its sovereign debt.

The background of the slide is a grayscale photograph of modern buildings. On the left, a portion of a white building with rectangular windows is visible. On the right, a taller building with a curved facade and large glass panels is shown, with some palm trees in the foreground. A solid dark blue vertical band runs down the center of the image, serving as a background for the text.

09

Other Types of Long-term Debt Securities



Structured Notes: High Returns, Hidden Risks

Structured notes are debt securities where interest and principal payments depend on specified market conditions—tied to Treasury bond indices, stock indices, or currencies. While they became popular in the 1990s for their high return potential, they carry substantial risks that aren't always apparent.

The most dramatic example: Orange County's 1994 bankruptcy. The portfolio manager invested over \$7 billion in structured notes betting interest rates would decline. When rates rose instead, values plummeted. The manager borrowed more to invest further, compounding losses and leading to bankruptcy.

In 2018, Wells Fargo paid \$4.1 million to settle SEC charges for failing to disclose structured note risks and fees. Even AAA-rated structured notes proved risky—credit ratings have been frequently revised downward.

What They Are

Debt securities with payments tied to market conditions like bond prices, stock indices, or currencies

The Appeal

High return potential and ability to bet indirectly on restricted markets

The Danger

Complex risk assessment, misleading credit ratings, and potential for catastrophic losses



Alternative Debt Securities: ETNs and Auction-Rate Securities

Exchange-Traded Notes (ETNs)

ETNs are debt instruments promising returns based on a specific debt index performance after fees. With maturities of 10 to 30 years, they're issued by firms like Goldman Sachs and Morgan Stanley but aren't secured by assets, creating default risk.

Because ETNs can contain commodities and foreign currencies, they avoid mutual fund regulations and can use leverage—borrowed funds that magnify both gains and losses. This creates higher potential returns but significantly greater risk.

Auction-Rate Securities

Used since the 1980s by municipalities and student loan organizations, these securities allow long-term borrowing through short-term investor commitments. Every 7 to 35 days, securities are auctioned at new reset rates.

The \$330 billion market collapsed in 2008 when financial institutions couldn't find buyers and investments became frozen. Values declined substantially, and investors claimed they weren't informed of liquidity risks. Despite pressure from the SEC forcing some buybacks at face value, over \$40 billion in pre-crisis auction-rate notes remains outstanding with most auctions failing due to lack of investor interest.



Summary

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Summary of Bond Markets

Bond markets facilitate diverse funding activities through various participants and bond types. Liquidity, credit ratings, and market stability are vital for wise investment and financial system health.

- Bond markets are essential for funding various activities, from government infrastructure projects and residential mortgages to corporate expansions.
- Key participants include major banks managing bond issuances, insurance companies investing premiums, and pension funds securing retirement incomes.
- Bonds come in diverse forms, such as Indonesian Government Bonds (SUN) which are considered low risk, corporate bonds issued by Indonesian companies (varying risk), and Sharia-compliant bonds (Sukuk).
- Their liquidity depends on trading volume, and features like call provisions or convertibility add complexity.
- At any given time, yields differ due to factors like credit ratings, maturity dates, and prevailing market conditions. For example, Indonesian bonds might offer higher yields than US Treasury bonds to compensate for perceived higher risk.
- For wise investment, investors must carefully evaluate these risks and potential returns in the Indonesian bond market.
- Trust in financial institutions and the stability of bond markets are crucial for the overall health and functionality of the financial system, as seen in Indonesia's efforts to maintain financial stability during economic fluctuations.





Thank you